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Computer Aided Molecular Design: A Novel Method for Optimal Solvent Selection

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ABSTRACT

The task of selecting a solvent or solvent mixture with desirable combination of physical properties, to meet the needs of specific applications, has largely been tackled using a combination of heuristics and costly experimental studies. This material selection problem is here formulated as the combinatorial molecular design problem of choosing a set of structural groups making up a target molecule with the desired properties as predicted by available group contribution techniques. A novel mixed-integer nonlinear programming (MINLP) technique is used to solve the problem yielding compounds with optimum value of an appropriate performance index, subject to material balances, process and design limitations and feasibility of molecular structures. The strategy is applied with excellent results to solvent design examples for liquid-liquid extraction and multicomponent gas absorption using varying combinations of objective functions and constraints to reflect directly a multiplicity of operational objectives.

INTRODUCTION

The choice of an appropriate solvent for a gas absorption or liquid extraction operation is often very significant in ensuring the viability and performance of many industrial processes. The primary requirement is for the solvent to be highly selective in separating desired products from other components in a feed mixture. Other solvent characteristics such as viscosity, capacity and toxicity must also be considered and the final choice typically involves some compromise as no single compound can usually demonstrate the best performance in all desirable characteristics.

This solvent selection problem has hitherto been tackled using a combination of fundamental physico-chemical knowledge, heuristics and costly experimental studies. More recently, techniques have been proposed to make advantageous use of the group contribution concept in performing a computer aided molecular design to identify new compounds, thereby transferring the solvent search from the laboratory to the desktop (Brignole *et al.*, 1986; Joback and Stephanopoulos, 1989; Nielsen and Gani, 1990; Macchietto *et al.*, 1990).

Computer aided molecular design is essentially the inverse of property prediction by group contribution. Given a set of desirable properties, it is proposed to find a combination of structural groups, and hence a molecular structure, satisfying the property specifications. Potential feasible molecules may also be ranked according to a suitably chosen performance measure.

Brignole *et al.* (1986), Joback and Stephanopoulos (1989), Nielsen and Gani (1990) use different techniques involving the explicit generation and testing of feasible molecular structures, sometimes using specific heuristics or knowledge-based techniques to reduce the enormous search space.

The approach employed by Macchietto *et al.* (1990) is to evaluate all feasible molecular structures implicitly within the framework of a constrained optimisation formulation. The design of a "best" solvent for a specific application is achieved by using thermodynamic and group contribution models to evaluate directly a quantitative measure of performance related to molecular structure (say, the selectivity of a solvent in a liquid-liquid equilibrium stage). This performance measure is adopted as an objective function for optimisation with respect to the number and type of the structural groups constituting the molecule, subject to constraints of molecular structure feasibility and other relevant engineering criteria.

The optimisation formulation permits a great deal of flexibility as various options can be explored for the same problem using different combinations of design variables, objective function and constraints to exploit the degrees of freedom inherent in an optimisation formulation whilst reflecting directly and explicitly the multiplicity of objectives involved in molecular design. The search can also be targeted at a specific class of compounds such as aromatics or alcohols by judicious formulation of the constraints.

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Feasibility of the solvent structure requires that the number of each structural group type present in the molecule assumes integer values. Hence the molecular design problem is posed as a mixed-integer nonlinear programming (MINLP) problem. Previous work using continuous optimisation has demonstrated the viability of the optimisation approach but dealt with the integrality of the structural group numbers by suitable rounding or addition of integer forcing constraints after finding a (relaxed) solution with non integer number of groups (Macchietto *et al., 1990)*. The solvents identified could not be guaranteed as optimal as the MINLP was not solved rigourously. Here, a general MINLP solver, based on the augmented-penalty/outer-approximation (AP/OA) algorithm proposed by Viswanathan and Grossmann (1990), has been developed for this purpose.

The incorporation of the new MINLP method in the molecular design guarantees optimality of the identified solvent structures. By starting off the design procedure using industrial solvents as initial "guesses", rapid location of optimal solvents, demonstrating better combination of desirable properties than those currently used, is usually achieved in a few iteration steps.

The mathematical formulation of the molecular design problem as a nonlinear optimisation is presented in the next section followed by a very brief description of the integer programming algorithm employed in solving the problem. The feasibility of the approach is demonstrated by applications in multicomponent gas absorption and liquid extraction.

MATHEMATICAL FORMULATION

Given:

1. A specific application process requiring molecule(s) with desirable properties (e.g. a single liquid-liquid equilibrium extraction stage),
2. Relevant operating conditions,
3. Appropriate design criteria,
4. A quantifiable measure of performance,
5. Predictive techniques for the estimation of the physical and chemical properties, and
6. A set of constituent structural groups (e.g. CH_3, CH_2O ...) with associated group contribution parameters,

the design problem is to use the available groups as building blocks to assemble a feasible molecule with optimal value of the performance measure and possessing characteristics consistent with the specified design criteria. This is posed mathematically as the following optimisation problem:

P1 $\max_{\boldsymbol{n},\boldsymbol{x}} p(\boldsymbol{n},\boldsymbol{x})$ (performance measure)

subject to:

$$\left.\begin{aligned} \boldsymbol{h}(\boldsymbol{n},\boldsymbol{x}) &= 0 \\ \boldsymbol{g}(\boldsymbol{n},\boldsymbol{x}) &\le 0 \end{aligned}\right\} \text{ (process equations, feasibility criteria, design specifications)}$$

Here, p is the appropriately defined performance measure (e.g. solute recovery, solvent selectivity, or a combination of these), $\boldsymbol{n}$ is a vector of integer variables characterising the molecular configuration with scalar variables n_j indicating the number of structural groups of type j ($j \in I$, a set of allowable groups) in the molecule, and (optionally) the number of solvent components, equilibrium stages, etc. $\boldsymbol{x}$ is a vector of continuous variables representing the equilibrium phase compositions and, optionally, other variables such as the solvent to feed ratio, temperature and/or pressure, etc. $\boldsymbol{h}$ and $\boldsymbol{g}$ are general equality and inequality constraints, respectively, describing the material and energy balance equations, physical property relations, structural feasibility criteria, logical constraints and other design specifications.

This general formulation of the molecular design problem allows for the possibility of objective and constraint functions exhibiting nonlinearity in both continuous and discrete decision variables. The functions may also be defined as implicit or explicit in the decision variables and may be evaluated by analytical or numerical techniques.

Performance Measure and Process Constraints

The performance measure and process constraints will be dependent on the actual requirements of a particular problem. Here we consider a single equilibrium stage as a representative process. The solvent properties selectivity, capacity and solvent loss were used here in formulating the process constraints. The selectivity, Sel_{ij}, of the solvent for a component i relative to component j and the solvent capacity, Cap_i, for a component i are defined as:

$$Sel_{ij} = \frac{x_i^S \cdot x_j^F}{x_j^S \cdot x_i^F}; \quad Cap_i = \frac{x_i^S}{x_i^F}$$

where x_i^S, x_i^F represent the equilibrium compositions of component i in the solvent-rich and solvent-lean phases respectively. The solvent loss is defined as the fraction of solvent feed at equilibrium in the solvent-lean phase. Mass balances and equilibrium equations are included in $\boldsymbol{h}(\boldsymbol{n},\boldsymbol{x})$ as process constraints.

Structural Feasibility Constraints

By classification of the different structural groups on the basis of their valencies (number of free attachments), the octet rule provides a simple relation for structural feasibility of a collection of groups;

$$\sum_j (2 - v_j) n_j = 2m \tag{1}$$

where n_j, v_j are the number and valency, respectively, of groups of type j and m = 1, 0 or -1 for acyclic, monocyclic and bicyclic groups, respectively.

In previous work (Macchietto *et al.*, 1990) the above constraint could only be applied for each class of solvent by performing the molecular design three times with a fixed value of m, having selected apriori the type of optimal molecule desired. The current formulation allows the simultaneous consideration of all the possibilities by considering m as an integer decision variable with a lower limit of -1 and upper limit of 1.

Structural feasibility further requires that two adjacent groups in a molecule may not be linked by more than one bond. This implies that for each structural group present, there must be at least as many *other* groups as available attachments. Given n_j groups of type j with valency v_j, a total of $n_j(v_j - 2) + 2$ attachments are available for bonding. Thus;

$$\sum_{i \neq j} n_i \geq n_j(v_j - 2) + 2 \quad \text{for all groups } j$$

$$\Rightarrow \sum_i n_i \geq n_j(v_j - 1) + 2 \quad \text{for all groups } j \tag{2}$$

Additional constraints may be placed on the number, n_j, of groups of a particular type j to keep it within lower and upper bounds n_j^L and n_j^U, respectively;

$$n_j^L \leq n_j \leq n_j^U \tag{3}$$

Finally a constraint may be imposed on the total number of structural groups making up a molecule. This is necessary as many group contribution methods become less accurate as the number of structural groups in a molecule becomes larger. Thus;

$$1 \leq \sum_j n_j \leq nmax \tag{4}$$

where *nmax* is the maximum number of structural groups allowed to make up a feasible solvent structure and has been set to 10 for all the examples presented in this work. Other constraints limiting potential combinations of groups are given in Macchietto *et al.* (1990).

Evaluation of Physical and Equilibrium Properties

Physical and equilibrium properties required in the design are obtained by a combination of group contribution and empirical correlations. The functional relationships for the properties used in this work are summarised in Table 1. The cited references can be consulted for more detailed information on each prediction method.

TABLE 1. Functional Relationships of Physical properties used in the design.

Property	Model	Functionality
Normal Boiling Point	Lai *et al. (1987)*	$T_b = f(\boldsymbol{n})$
Critical Properties	Joback (1984)	$T_c, P_c = f(\boldsymbol{n})$
Activity Coefficients	UNIFAC (Larsen *et al.* (1987)	$\gamma_i = f(\boldsymbol{n},\boldsymbol{x})$
Fugacity Coefficients	UNIWAALS (Gani *et al.* (1989)	$\phi_i = f(\boldsymbol{n},\boldsymbol{x},T_c,P_c)$
Vapour Pressure	Riedel (Reid *et al.* (1987)	$P_{vp} = f(T_b,T_c,P_c)$
Heat Capacity	Joback-Rowlinson (Reid *et al.* (1987)	$C_p = f(\boldsymbol{n},\boldsymbol{x},T_c)$

Solution Procedure

The implementation proceeds with an initial estimate, $\boldsymbol{n}^0$, of the optimum solvent. Process variables are either specified or given as initial estimates depending on the molecular design goals. The corresponding equilibrium molefractions, in the vector $\boldsymbol{x}^0$, are determined by phase equilibrium calculations using the appropriate group contribution-based methods in Table 1. The objective function and constraints are then evaluated for the solvent process being examined. Derivative information, relating objective and constraint functions to all optimisation variables, is employed in guiding the optimisation. These are calculated by a combination of numerical and, where possible, analytical derivation.

Initially, the optimisation problem P1 is solved in its relaxed form, where all optimisation variables are considered to have continuous values. A standard program for nonlinear programming (NLP) implementing a successive quadratic programming method (Chen, 1988) is used for this purpose. The solution to the relaxed problem corresponds to the required optimum solvent provided all variables in the $\boldsymbol{n}$ vector turn out to be integers. Otherwise integer estimates of the solution are obtained using a branch and bound technique and solving a mixed-integer linear program (MILP) in which the nonlinear objective and constraint functions are linearised at the current continuous solution. The optimisation variables for this MILP are the original variables and additional non-negative slack variables which allow the linearised constraints to be violated at the expense of being penalised in a modified objective function. An NLP subproblem of the original problem P1 is now solved with the integer variables determined by the solution of the preceding MILP, to give new values of the continuous variables and a new point for linearising the nonlinear functions. A finite sequence of MILP master problems and NLP subproblems is then solved until there is a decrease in the value of the objective function for successive feasible NLP subproblems. An integer cut is also included in each MILP to exclude previously identified integer combinations from the search space. This procedure is illustrated schematically in Figure 1.

The algorithm makes some new contributions to the solution of MINLP problems. General integer variables are explicitly handled as opposed to available packages which require a problem reformulation in the space of 0-1 variables. The original outer approximation methods were limited to problems involving separable functions in which all nonlinearities are confined to the continuous variables with integer variables occurring linearly. The present algorithm handles general nonlinear functions and constraints. It has been shown to locate the global optimum very efficiently in all cases for a set of mathematical test problems and is presented in more detail elsewhere (Odele, 1992). This rigourous MINLP procedure completely resolves previous problems in obtaining an integer solution for the numbers of groups.

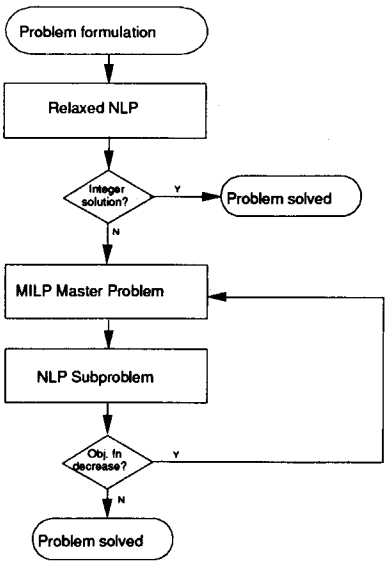


Figure 1. Flowchart of the algorithm used for the solution of the MINLP molecular design problem

MOLECULAR DESIGN EXAMPLES

The molecular design strategy is illustrated for applications in liquid extraction and gas absorption. A single equilibrium stage is taken in each case as a representative process.

Solvent Design for Gas Absorption

In a gas absorption system, solvent choice, process configuration, equipment and operating conditions are all critical variables to be considered. The solvent selection aspect is considered on the basis that a process arrangement and equipment have already been proposed. Once the solvent has been selected, it may then be possible to adapt the configuration, equipment and conditions to take further advantage of specific solvent properties.

The objective here is to design solvents for the efficient absorption of a multicomponent feed representing a typical industrial sour gas for sweetening (methane, carbon dioxide and hydrogen sulphide with composition 80, 15, 5 mole percent, respectively). The absorption system is at 300 K and 30 atm. The feed to the absorber consists of the solvent and feed gas in equimolar amounts.

The following structural groups, for which a complete set of the necessary interaction parameters is available, were selected as candidates for the optimal molecular structure; I = {-CH_3, -CH_2-, >CH-, >C<, -OH, CH_3OH, H_2O, -$COCH_3$, -$COCH_2$-, -CHO, CH_3COO-, -CH_2COO-, CH_3O-, -CH_2O-, >CH-O- }. Other groups were not included due to the current lack of interaction parameters (e.g. parameters for interaction between amines and the gases).

In this example, we first required the identification of a solvent with the greatest selectivity for absorption of hydrogen sulphide relative to carbon dioxide, with the optimal solvent allowed to consist of at most one non-hydrocarbon functional group. The design specifications and iteration sequence are shown in Table 2, starting with methanol as the initial solvent. At the first iteration, corresponding to the solution of the relaxed problem, an hypothetical compound with fractional numbers of groups was obtained. The optimal solvent structure, isopropyl methyl ether is identified at the fourth major iteration[1] after which the objective function decreases. The other compounds identified along the path also demonstrate great potential for use as selective solvents. The ether group, CH_3O, which appears in the best solutions, is clearly the functional group with the best potential for highly selective absorption.

TABLE 2. Optimisation sequence for selective absorption; objective function = selectivity

Itn.	Groups in molecular structure	Selectivity	Capacity	Solvent loss(%)
0	CH_3OH	3.6	0.26	0.1
1	$(CH_3)_{0.3}$ $(CH_2)_{0.3}$ $(CH_3OH)_{0.3}$ CH_3O	20.9	0.34	8.5
2	CH_3 CH_3O	3.0	0.01	95.0
3	$(CH_3)_2$ CH_2 CH CH_3O	10.4	0.64	3.0
4	**$(CH_3)_2$ CH CH_3O**	**11.8**	0.58	7.0
5	CH_3 $(CH_2)_2$ CH_3O	11.5	0.69	1.8

Design specifications: capacity ≥ 0.1, functional groups ≤ 1; solvent loss $\leq 10\%$, acyclic molecule ($m = 1$)

The maximum solvent loss constraint was then relaxed from a maximum of 10% to 20% and the optimisation performed as before. Once again the optimisation converged after a total of only 5 major iterations with the identification of a different optimal solvent with improved selectivity (Table 3, iteration 4). This simple experiment demonstrates the flexibility of the optimisation approach in handling directly changes in design specifications and in reflecting the effects of such changes on the performance index.

The solvent design problem described in Table 3 had been previously considered by Odele and Macchietto (1990) using a continuous optimisation algorithm coupled with integer forcing constraints. The compound (propyl methyl ether) found at iteration 3 in Table 3 was identified there as the best integer solution. However, the use of a more powerful optimisation algorithm in the present work has led to the identification of a significantly more selective solvent, ethyl methyl ether.

1 N major iterations require the solution of N NLP subproblems and N-1 master problems.

TABLE 3. Optimisation sequence in solvent design for gas absorption; objective function = selectivity

Itn.	Groups in molecular structure	Selectivity	Capacity	Solvent loss(%)
0	$(CH_3)_2$ $(CH_2)_4$	4.2	0.44	2.9
1	$(CH_3)_{0.3}$ $(CH_2)_{0.3}$ $(CH_3OH)_{0.3}$ CH_3O	20.9	0.34	8.5
2	CH_3 CH_3O	3.0	0.01	95.0
3	CH_3 $(CH_2)_2$ CH_3O	11.5	0.69	1.8
4	**(CH_3) CH_2 CH_3O**	**13.9**	0.46	18.0
5	$(CH_3)_2$ CH_2O	11.1	0.46	17.9

Design specifications: solvent loss ≤ 20%, capacity ≥ 0.1, functional groups ≤ 1, ($m = 1$)

A molecular design with temperature optimisation was also performed using the same combination of objective function and constraints as given in Table 2 but in addition allowing the absorption temperature to vary between 290 K and 310 K. The same solvent as identified earlier, isopropyl methyl ether gave the highest selectivity of 12.4 at 308 K with a reduced capacity of 0.49, the solvent loss constraint being active at this optimum. The same design problem was then solved using a different objective function ($p = Sel*Cap - 100*$Solvent loss) and a different optimal solvent, propyl methyl ether, was identified with the temperature at the lower bound. These results with temperature optimisation show that lower solvent losses and higher capacity are favoured by a reduction in the absorption temperature as expected. The selectivities for the identified compounds however increase with temperature, yielding an interesting optimisation problem.

The effect of additional functional groups in the solvent molecule is demonstrated by the results shown in Tables 4 and 5 in which up to 2 functional groups are allowed. In the first case, the algorithm is let free to choose amongst acyclic, monocyclic and bicyclic compounds. The nonlinear relaxation of the problem produces a molecule for which m is 0.18. The optimal structure is cyclic and is identified at the second step, however no existing compound is known to possess such a structure. The molecular design was then restricted to identification of acyclic structures only. Slightly reduced selectivities of 33.7 and 33.5 are now obtained for the optimal solvent (dimethyl acetal acetaldehyde) and ethylene glycol dimethyl ether (iteration 3 of Table 5), respectively. These compare with the value of 11.8 for the optimal solvent in the equivalent problem with only one functional group allowed (Table 2).

TABLE 4. Iteration sequence for selective gas absorption; objective function = selectivity

Itn.	Groups in molecular structure	Selectivity	Capacity	Solvent loss(%)
0	CH_3 $(CH_2)_3$ (CH_3O)	9.5	0.62	4.3
1	$(CH_2)_{0.2}$ $(CH)_{0.7}$ $(C)_{0.5}$ $(CH_3O)_2$	74.6	0.30	9.9
2	**(CH_2) (CH) (CH_3O) (CH_2O)**	**39.6**	0.42	6.7
3	(CH_3) (CH) $(CH_3O)_2$	33.7	0.47	5.2

Design specifications: solvent loss ≤ 10%, capacity ≥ 0.1, functional groups ≤ 2 , $-1 \le m \le 1$

TABLE 5. Solvent Design for gas absorption; objective function = selectivity

Itn.	Groups in molecular structure	Selectivity	Capacity	Solvent loss(%)
0	CH_3 $(CH_2)_4$ CH_3CO	0.6	0.39	0.5
1	$(CH3)_{0.6}$ $(CH)_{0.6}$ $(CH_3O)_2$	47.9	0.37	9.0
2	**(CH_3) CH $(CH_3O)_2$**	**33.7**	0.47	5.2
3	$(CH_2)_2$ $(CH_3O)_2$	33.5	0.47	5.5

Design specifications: solvent loss ≤10%, functional groups ≤ 2, capacity ≥ 0.1, $m = 1$

These results lead us to infer that the best combination of high selectivity, low solvent losses and high capacity desired for the given absorption system is induced by the presence of the ether groups in the solvent molecule.

The molecular design problems were solved using a variety of compounds as initial "guesses" with the same relaxed and integer optima obtained in all cases, suggesting that the identified molecular structures indeed represent the global optimum for the chosen set of constituent groups and constraints.

Solvent Design for Liquid Extraction

The solvent design strategy is demonstrated by application to two important industrial liquid extraction problems, namely the dehydration of acetic acid and ethanol solutions. Again, a single liquid-liquid equilibrium stage is taken as the reference process.

Solvent Design for Acetic acid extraction

Solvent design examples for the extraction of acetic acid from water have been presented by Macchietto *et al.* (1990). The optimal solvent in this work is to be made up of the same set of groups I = {CH_3, CH_2, CH, CH_3CO, COOH, CHO, CH_3COO, CH_3O, CH_2O, CH-O} as used in their original problem. The molecular design of a high boiling solvent is repeated here using the new optimisation algorithm. The feeds are water, acetic acid and solvent in a molar ratio of 6:1:3 at 298 K and 1 atm. The design pathway to the optimum solvent, starting from the industrially used methyl isobutyl ketone as the initial guess, is presented in Table 6. The optimal solvent located at the fourth major iteration is the same as that identified in the earlier work (tetramethyl hexane) with a performance index of 421.9, giving an integer gap of 1.8% with respect to the relaxed optimum. In addition to this optimum solvent, three new alternative solvents with desirable combinations of selectivity and capacity were identified along the optimisation sequence. These compounds were also identified as having excellent prospects for the acetic acid extraction by Nielsen and Gani (1990).

TABLE 6. Design of solvent for acetic acid extraction; objective function = selectivity * capacity

Itn.	Groups in molecular structure	Selectivity	Capacity	p = Sel * Cap
0	CH_3 $(CH_2)_3$ CH_3CO	11.2	1.80	20.2
1	$(CH_3)_{3.5}$ $(CH_2)_5$ $(CH)_{1.5}$	524.5	0.82	429.3
2	$(CH_3)_3$ $(CH_2)_5$ CH	516.8	0.78	403.1
3	$(CH_3)_4$ $(CH_2)_3$ $(CH)_2$	522.5	0.80	420.2
4	**$(CH_3)_6$ $(CH)_4$**	**518.1**	0.81	421.9
5	$(CH_3)_5$ $(CH_2)_2$ $(CH)_3$	518.0	0.81	421.6

Design specification: solvent boiling point ≥ acetic acid boiling point + 20 K , m = 1

Solvent Design for Ethanol Extraction

The ternary system water-ethanol-solvent in a molar ratio of 6:1:3 at 298 K and 1 atm is investigated with the objective of identifying an optimum solvent to preferentially extract the ethanol. The same ten structural groups used in the acetic acid extraction are used to make up the solvent molecule, with the exception of the carboxylic acid group (COOH) which is replaced by the hydroxyl group (OH) to prevent esterification. Results are indicated in Table 7, starting with heptane as the initial solvent and terminating to an optimal structure corresponding to 2,3,4-trimethylpentane.

TABLE 7. Design of solvent for ethanol extraction; objective function = selectivity * capacity

Itn.	Groups in molecular structure	Selectivity	Capacity	p = Sel * Cap
0	$(CH_3)_2$ $(CH_2)_5$	138.7	0.19	26.4
1	$(CH_3)_{4.3}$ $(CH_2)_{1.4}$ $(CH)_{2.3}$	147.0	0.19	27.4
2	**$(CH_3)_5$ $(CH)_3$**	**157.4**	0.17	27.4
3	$(CH_3)_2$ $(CH_2)_6$	119.6	0.22	26.6

Design specifications: solvent boiling point ≥ ethanol boiling point + 20 K, m = 1

CONCLUSIONS AND SIGNIFICANCE

This paper has demonstrated that the group contribution based molecular design of solvents for gas absorption and liquid extraction processes is greatly enhanced by the availability of the new robust mixed-integer nonlinear programming implementation to solve the resulting optimisation problem. The main advantage of the method is the ability to incorporate directly in the search, very general engineering constraints and objectives.

Optimum solvents have been identified in all cases very efficiently and reliably in under six major iterations of the outer approximation algorithm with each major iteration using an average of 50 seconds of CPU time on a SUN Sparcstation. The linearity of all the structural feasibility constraints further ensures that each compound corresponding to an intermediate solution at each major iteration is structurally feasible and hence a potential alternative to the optimal solvent should this be found to be unsuitable (for reasons not reflected in the problem constraints) after subsequent evaluation.

The molecules displaying the best potential as regards the chosen performance criteria are observed to consist mainly of a specific subset of the groups which are the most conducive for the specific application. For the specific examples considered, the ether and alkyl groups are best constituents for gas absorption of acid gases from industrial gas streams, while the alkyl groups demonstrate great potential for use in the dehydration of acetic acid and ethanol solutions. These results are consistent with industrial practice.

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